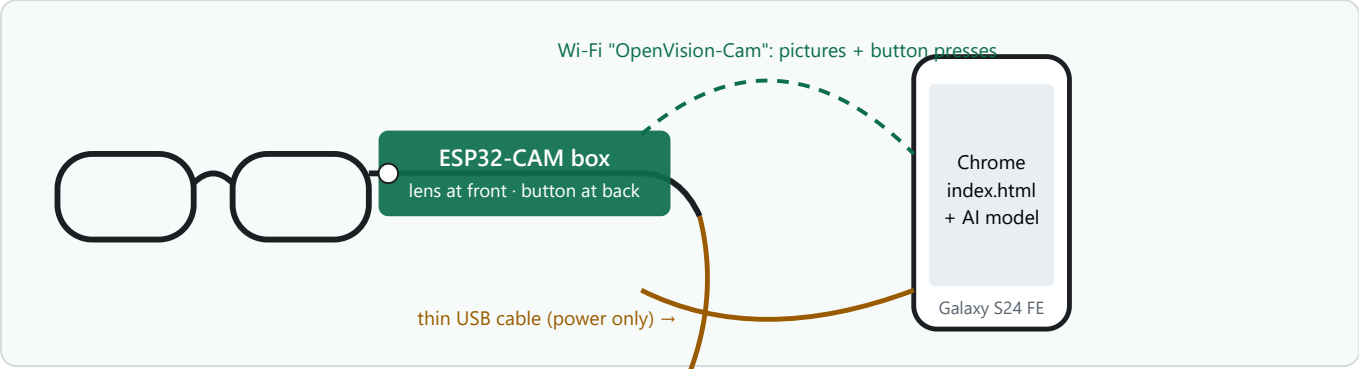
Build the OpenVision glasses camera

A Wi-Fi camera with a push button that clips onto ordinary glasses. It sends what you see to the AI model running in Chrome on your Galaxy S24 FE. **Tap** the button and the phone describes the scene out loud. **Hold** it and you can ask a question.

≈ 1.1–1.5 M T	4–6 h	≈ 18 g	66×31×13 mm
parts cost (Nov 2026, Iran)	work, plus print time	on one arm of the glasses	camera box · fits any glasses



How it works in one paragraph

The ESP32-CAM board (a small Wi-Fi chip plus a 2-megapixel camera) sits in a 3D-printed box on the outside of the temple arm, just behind the hinge. It makes its own Wi-Fi network, `OpenVision-Cam`. The phone joins that network, and the OpenVision page in Chrome fetches a picture whenever you press the button. The board gets power from the phone through a thin USB cable that runs along the arm and behind your ear. Everything stays offline on the phone.

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> **No calipers needed.** The box is sized from the official Ai-Thinker datasheet (board 27 × 40.5 × 4.5 mm ±0.2), with room to spare. It's held on by two straps that wrap around the box and the temple arm, so it fits almost any glasses (arms about 1.5–8 mm thick and 2–15 mm tall, metal or plastic). A ruler with mm marks is enough for the few checks.

1 · Shopping list

Prices are what Iranian shops listed in November 2026. They change often, so check before you pay. Quantities include a spare where parts are cheap and easy to break.

#	Part (search term)	Qty	Exact specification	Where · ≈ price
1	**ESP32-CAM + ESP32-CAM-MB** «ماژول ESP32-CAM با برد پروگرامر»	1 set	AI-Thinker ESP32-CAM, **OV2640** camera, 4 MB PSRAM. The MB is the small USB programmer board it plugs into (CH340 chip, micro-USB). Buy the bundle. **Ask for OV2640.** Boards sold with a **"RHYX-M21-45" (GC2145)** camera also work with this firmware, which detects them automatically, but each picture takes about 0.3 s longer because that camera can't make JPEG itself.	**Idea Electronic, Tehran** (Jomhouri St., corner of Si-Tir, Forghani bldg., 1st floor, No. 10), shop.ideaelec.com: board «ESP32 cam» OV2640 1,499,000 T + programmer shield (micro-USB, CH340G) 272,428 T. Both in stock on 2026-11-04. Other options: ECA has only the RHYX-M21 version in stock (2.26M T + 284k shield); Master Micro (Ardabil) sells an OV2640 board for 685k T, sent by post.
2	Tactile push button «تک سوییچ ۶×۶ ارتفاع ۷»	3	6 × 6 mm, **7 mm tall** (body + plunger), 4 through-hole legs. A soft or quiet type is nicer.	any electronics shop · small change
3	Electrolytic capacitor	2	**220 µF or 330 µF, 10 V or 16 V**, size no bigger than **6.3 mm diameter × 11 mm** (it has to fit the box).	small change
4	Thin USB cable, USB-A plug	1	USB-A male to anything, **1.2–1.5 m**, as thin and flexible as you can find (≈3 mm). You'll cut off the other end. A cheap charging cable is fine as long as it has red and black wires inside.	small change
5	OTG adapter	1	**USB-C male → USB-A female, labelled "OTG"**. This is how the phone powers the camera. A plain "charging adapter" may not work.	mobile accessory shop
6	Hook-up wire	30 cm	Silicone wire, **28 AWG** (or 26 AWG), two colours.	small change
7	Heat-shrink tubing	20 cm	1.5 mm and 4 mm diameter.	small change

8	**Straps for the box** «بست کمربندی ۱۵۰ میلی‌متری»	10	Zip-ties **150 mm long, 2.5–3.6 mm wide**: 2 to hold the box, 3–4 to hold the cable along the arm, the rest spare. *Nicer:* **reusable / releasable zip-ties** no wider than 4.8 mm («بست کمربندی بازشو»), or **silicone O-rings** about 30 mm inside diameter, 3 mm thick, which slip on and off without tools.	hardware / electrical shop · small change
8b	Anti-slip pad	1	**1 mm EVA foam or a strip of rubber** (an old bicycle inner tube is ideal), about 12 × 60 mm. Goes between the box and the arm so it can't slide or scratch the frame.	stationery shop · small change
9	PETG filament	≈ 20 g	Any colour; black hides the box best. No printer? Send the STL files to a print service («پرینت سه بعدی»).	or a print service
10	Glue + tape	–	Hot-glue stick, a drop of CA (super) glue, 1 mm double-sided foam tape.	stationery shop
11	*Optional:* OV2640 with a long ribbon	1	OV2640 module with a **75 mm** ribbon. Only needed if the stock ribbon can't bend the lens to the front of the box (step 6.3).	«ماژول دوربین OV2640 کابل بلند»

> You do **not** need: an Arduino Uno/Nano, a battery, a charger board, a boost converter, or the "pocket hub". The ESP32 chip on the camera board is the microcontroller, and the phone supplies the power.

2 · Tools and software

Tools

- ▢ **Ruler** with mm marks (calipers are *not* needed)
- ▢ Soldering iron with a fine tip, 0.5–0.8 mm solder, flux
- ▢ Flush cutters (for cutting the header pins)
- ▢ Wire stripper, tweezers
- ▢ Multimeter (voltage + continuity beep)
- ▢ Hot-glue gun, heat source for the heat-shrink
- ▢ 3D printer (0.4 mm nozzle) or a print service
- ▢ Micro-USB **data** cable (for the MB programmer)

Software (free)

- ▢ **Arduino IDE 2.x**: arduino.cc/en/software
- ▢ **OpenSCAD** 2021.01 or newer: openscad.org
- ▢ A slicer: PrusaSlicer / Cura / OrcaSlicer
- ▢ CH340 driver, only if Windows doesn't see the MB: search "CH341SER" (wch.cn)
- ▢ On the phone: **Termux** from **F-Droid** (not the old Play Store version), latest Chrome

Files from the project folder

- `firmware/openvision_cam/openvision_cam.ino`
- `hardware/glasses_cam_mount.scad`
- `index.html`

3 · Program the board and test it on the desk

Do this first, while the board still has its pins. If it doesn't work on the desk, it won't work on the glasses.

3.1 Install ESP32 support in Arduino IDE

1. Open Arduino IDE → **File** → **Preferences**.
2. In **Additional boards manager URLs**, paste:

   ```
   https://espressif.github.io/arduino-esp32/package_esp32_index.json
   ```

3. **Tools** → **Board** → **Boards Manager**, search `esp32`, and install **"esp32 by Espressif Systems"** (the newest version).

3.2 Open and edit the sketch

1. **File** → **Open** → `firmware/openvision_cam/openvision_cam.ino`.
2. Find `AP_PASS = "openvision123"` and change it to your own password, **at least 8 characters**. Write it down; you'll need it on the phone and for later updates.
3. Leave everything else as it is for now.

3.3 Connect and choose the exact settings

1. Fit the ESP32-CAM into the MB board (camera facing up, pins lined up), then plug the micro-USB cable into the MB and the PC.
2. Set the **Tools** menu exactly like this:

Tools menu item	Set to
Board	**esp32 → ESP32 Wrover Module**
PSRAM	**Enabled**
Partition Scheme	**Minimal SPIFFS (1.9MB APP with OTA / 190KB SPIFFS)**
Upload Speed	**460800** (use 115200 if uploads fail)
Port	the new **COM** port that appears when you plug in the MB

3. Click **Upload** (→). If it stops at `Connecting...` or says `Failed to connect`: **hold the IO0 button on the MB**, click Upload again, and let go when the percentage starts counting.
4. When you see `Hard resetting via RTS pin...`, press the **RST** button on the ESP32-CAM.
5. Open **Tools** → **Serial Monitor** at **115200 baud**. You should see:

   ```
   Wi-Fi "OpenVision-Cam" – camera at http://192.168.4.1
   ```

   If you see `Camera init failed` instead, the camera ribbon isn't fully in. Flip up the black latch, push the ribbon in (contacts facing the board), close the latch, and press RST.

3.4 Desk test with the phone

1. **Remove the clear protective film from the camera lens** (there's often a small tab on it).
2. On the phone: **Settings** → **Connections** → **Wi-Fi** → `OpenVision-Cam` → enter your password. If Android says "no internet", choose **Keep connection** / **Stay connected**.
3. Open Chrome and go to `http://192.168.4.1`. You should see "*OpenVision camera is running*" and a picture.

4. If the page won't load: turn off **mobile data** for a moment and try again (see Troubleshooting).
5. If the picture is upside down, set `FLIP_V = true` and `MIRROR_H = true` in the sketch and upload again. It's easier to fix now, while the pins are still on.

3.5 Test the button on the desk

1. Take a push button and touch two of its **diagonally opposite** legs to the **IO13** and **GND** pins (wires or jumper leads), with the Serial Monitor open.
2. A short press should print `button: down` then `button: tap`. A press of half a second or more should print `button: down`, `button: long`, `button: up`.

> **Use IO13, never IO12.** IO12 is a boot pin, and if it's pulled the wrong way the board won't start. The pins are labelled on the back of the board.

4 · Measure, edit the 3D model, print

4.1 Nothing to measure: the numbers already in the file

What	Model setting	Value	Where it comes from
Board length / height	`esp_l` / `esp_h`	40.5 / 27.0	Ai-Thinker datasheet (±0.2). The box adds 0.8 mm of clearance.
Board thickness	`esp_stack`	6.5	4.5 mm (datasheet) + 1 mm cut pin stubs + 1 mm foam
Camera module	`cam_sq`	9.0	OV2640 modules are about 8–8.5 mm; the seat is generous and the module is glued
Lens hole	`esp_lens_d`	8.0	Lens barrels are about 6–7.5 mm
Mounting	`mount_style`	strap	Fits any arm, so the arm size isn't needed

Ruler checks (when the board arrives): the board should be about 40–41 × 27 mm and no thicker than 6.5 mm once the pins are cut. The camera module should be no wider than 9 mm. If something is bigger, change that number in the Customizer before printing.

4.2 Set the model up in OpenSCAD

1. Open `hardware/glasses_cam_mount.scad` in OpenSCAD, then **Window** → **Customizer** (make sure it's ticked).
2. Check these settings (they're already the defaults):
   - **cam_type** = `esp32cam` · **mount_style** = `strap` · **preset** = `custom`
   - **side** = `right` (or `left`, for the arm you'll wear it on)
   - **tilt** = 10 (the camera points 10° down, which suits walking and reading)
3. Keep **part** = `preview` and press **F5**. The see-through bar is a temple arm, and the two see-through rings are the straps going around the box and the arm.

4.3 Export the two parts

For each of the two parts: set **part**, press **F6** (full render, wait for it to finish), then **File** → **Export** → **Export as STL**.

part =	Save as	Print how many	Orientation (already set by the file)
`mount`	ov_box.stl	1	Flat tray: the side that touches the arm goes on the bed, open side up
`lid`	ov_lid.stl	1 (+1 spare)	Flat, rim up
`cable_clip`	optional	0	Not needed: zip-ties or O-rings hold the cable (these clips are only for a known arm size)

4.4 Slicer settings

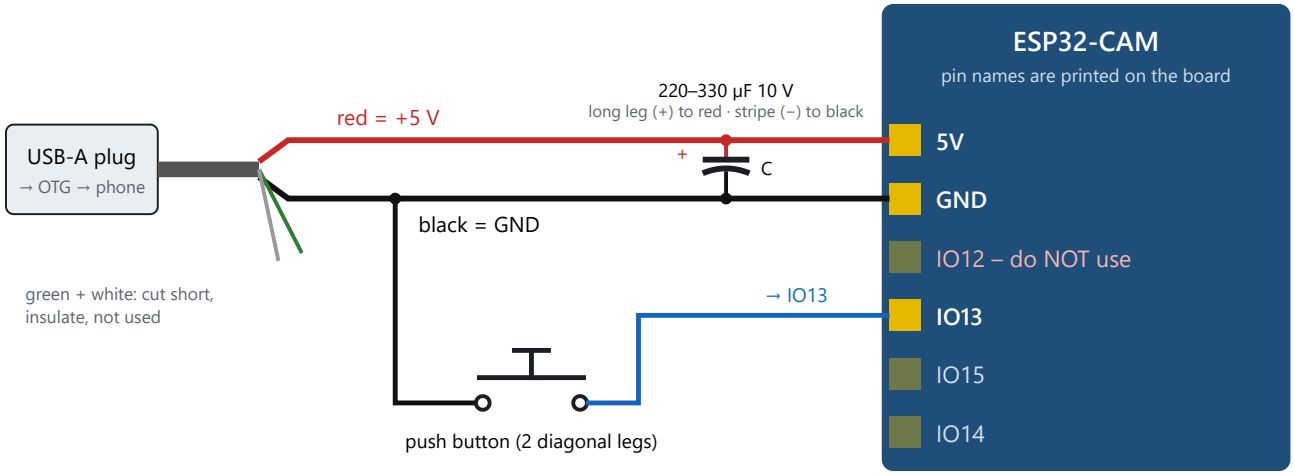
Setting	Value
Material	**PETG** (it copes with sun and body heat. PLA can soften in a hot car.)
Nozzle / layer height	0.4 mm / **0.2 mm**
Perimeters / top / bottom	**3** / 4 / 4

Infill	30 %, gyroid
Supports	**None** (the parts are designed to print without them)
Bed adhesion	None needed (both parts lie flat). Add a 3 mm brim if your PETG tends to lift.
Temperatures	whatever your PETG roll says (typically 230–245 °C nozzle, 75–85 °C bed)

4.5 Test-fit the print before any electronics go in

- ☐ The lid's rim drops into the box opening.
- ☐ A zip-tie (or O-ring) fits between each pair of fins on the top and bottom of the box.
- ☐ Hold the empty box against the outside of the arm, just behind the hinge, and wrap the two straps around box + arm (section 6.4). It shouldn't twist when you push the front.

5 · Wiring and soldering



Wiring diagram (not to scale). There are only three connections: 5 V, GND and the button on IO13. The capacitor sits across 5 V and GND close to the board.

5.1 Cut the header pins

1. Only do this **after the desk test passed** (section 3). From now on, updates go over Wi-Fi (section 9).
2. Take the board off the MB. With flush cutters, cut every pin **as short as you can** on both rows. Leave about 1 mm on **5V, GND and IO13** so you have something to solder to.
3. Brush away the metal bits. Hold the board up to a light and check that no two pins are bridged.
4. With a ruler, check the board is no thicker than **6.5 mm** now (the box is made for that). If it's thicker, file the pin stubs down.

5.2 Prepare and check the cable

1. Cut the cable so that, with the USB-A plug at one end, it's long enough to run from the hinge, along the arm, behind your ear and down to your pocket (usually **1.2 m**).
2. Strip 25 mm of the outer jacket. Inside you'll find red, black, green and white wires (sometimes a foil shield too, which you cut off).
3. **Check the colours with the multimeter:** plug the USB-A end into a phone charger and measure between red and black. You should read **+5.0 V** (±0.25), with red on the multimeter's red probe. Some cheap cables use odd colours, so trust the meter, not the colour.
4. Cut green and white back to 5 mm and cover them with heat-shrink.
5. Thread the cable in through the **exit slot at the back-bottom of the box** before you solder anything. Then pull a **2.5 mm zip-tie tight** around the cable about 8 mm from the end of the jacket and trim its tail. That's the strain relief: the zip-tie can't pass back through the slot.

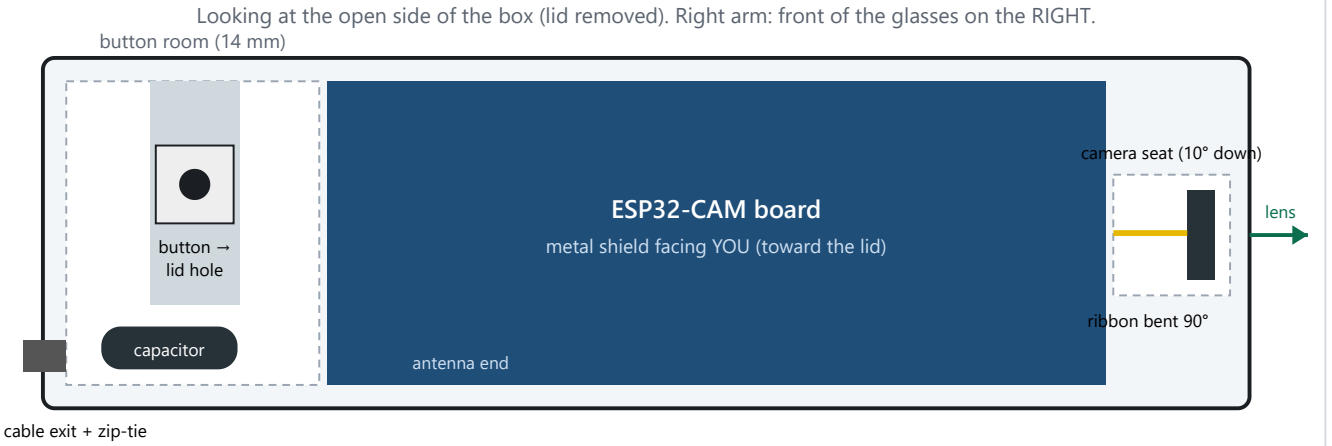
5.3 Solder (power off)

1. **Capacitor:** trim its legs to 6 mm. Solder **red** to the **long (+)** leg and **black** to the **striped (−)** leg. Put heat-shrink over each joint.
2. From the capacitor legs, run two 28 AWG wires about **35 mm** long: **+ → 5V pin**, **− → GND pin**.
3. **Button:** bend all 4 legs out flat. Solder two 28 AWG wires about **45 mm** long to **two diagonally opposite legs**. With the multimeter on continuity, check that it beeps **only while you press**.

4. One button wire goes to **IO13**. Join the other to the **GND** wire or GND pin (there's a second GND pin on the other row you can use too).
5. **Power test before closing:** plug the USB-A end into a charger. Within about 5 s the phone should see the `OpenVision-Cam` Wi-Fi network again.

> Never connect the 5 V wire to the **3V3** pin, and never reverse red and black. Either one can kill the board instantly.

6 · Assemble the box and fit it to the glasses



Layout inside the box. The board lies flat against the inner wall (the side toward your head), with its metal shield toward the lid (the vents). The camera ribbon bends forward into the angled seat behind the lens hole.

6.1 Board

1. Put a strip of 1 mm foam tape on the inner wall of the board area.
2. Lay the board in with the **antenna end toward the back** (the ear) and the **camera end toward the front**, metal shield facing out toward the lid.

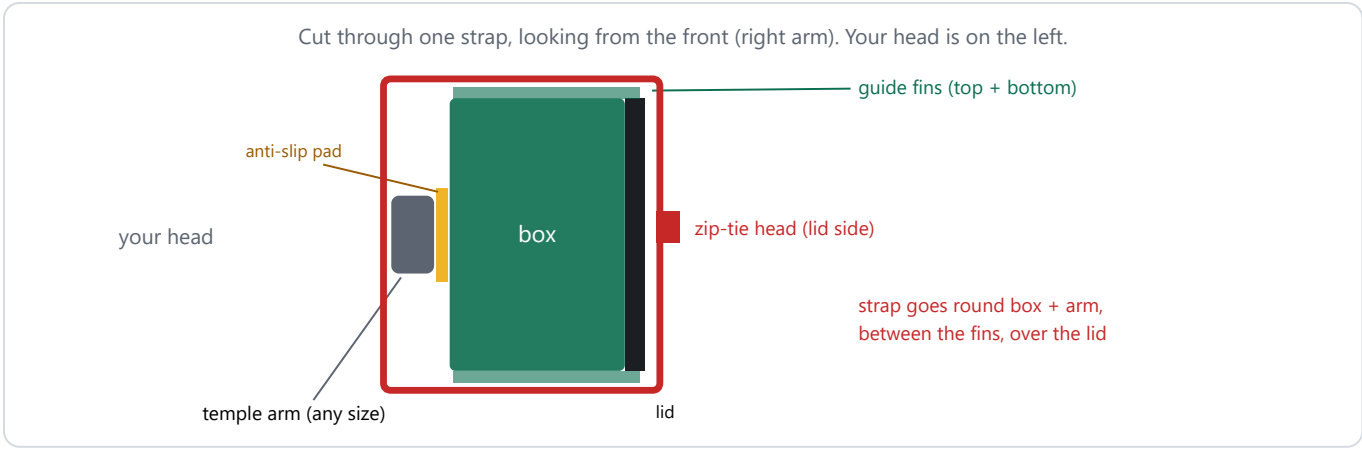
6.2 Button and capacitor

1. Sit the button on the **block** in the back chamber, plunger facing out toward the lid. Put the lid on loosely to check that the plunger pokes through its hole by about 1 mm. Then fix the button to the block with a dot of hot glue on its sides (not on the plunger).
2. Lay the capacitor in the channel **below** the block, with the cable and zip-tie behind it.

6.3 Camera

1. Gently bend the camera ribbon **90°** forward. Make it a smooth curve, never a sharp crease, and never bend right next to the connector.
2. Seat the camera module flat against the angled front face, with the lens in the hole. Before gluing, plug in power, open `http://192.168.4.1` and check the picture is upright and centred.
3. Fix it with **two small dots of CA glue** on the edges of the camera PCB. Keep glue well away from the lens.
4. If the ribbon won't reach without straining, use the 75 mm-ribbon OV2640 (shopping list #11). It plugs into the same connector.

6.4 Close and strap it to the glasses



The strap holds the box against the arm (with the pad between them) and also holds the lid on. Use two straps: one just behind the camera, one just in front of the button.

1. Press the lid on (rim inside). No glue needed, because the straps hold it.
2. Stick the **anti-slip pad** (foam or inner-tube rubber) onto the flat inner face of the box, the side with no opening.
3. Hold the box against the **outside** of the temple arm, **front edge about 5 mm behind the hinge**, lens pointing forward. Line the arm up with the **lens hole height**: the lens should sit level with your eye or slightly below.
4. Thread a zip-tie between the **front** pair of fins: over the top of the box, down the inner side of the arm, under the bottom, and back up over the lid. Put the zip-tie head on the lid side, where it can't touch your head. Pull it snug but don't crush the arm. Do the same at the **rear** fins.
5. Check: the box doesn't twist when you push it, and the glasses still fold. Trim the zip-tie tails flush with no sharp edge. (With O-rings: just stretch them over box + arm in the same two places.)
6. Run the cable back along the arm and hold it with **3–4 small zip-ties or O-rings**: mid-arm, just before the ear, and 1–2 on the curve behind the ear. Leave a small loop at the hinge so the arm can still fold.
7. Down to the phone: **USB-A plug → OTG adapter → phone**. The phone stays in your pocket.

7 · Set up the phone (Galaxy S24 FE)

> The page has to be opened from `http://localhost`. Chrome blocks an `https://` website from talking to the camera's plain `http://` address. Termux gives the phone its own small local web server.

7.1 Copy the app to the phone

1. Connect the phone to the PC with USB, create the folder `Download/OpenVision` on the phone, and copy `index.html` into it.

7.2 Termux (one time)

1. Install **F-Droid** from f-droid.org, then install **Termux** from F-Droid.
2. Open Termux and type these lines, pressing Enter after each:

   ```
   pkg update -y
   pkg install -y python
   termux-setup-storage
   ```

   Tap **Allow** when it asks for file access.

7.3 Start the app (every time)

1. In Termux:

   ```
   cd ~/storage/downloads/OpenVision
   python -m http.server 8080
   ```

   Pull down the Termux notification and tap **Acquire wakelock**, so Android doesn't stop it.
2. In Chrome open `http://localhost:8080/index.html`. Use ⋮ → **Add to Home screen** so it's one tap next time.

7.4 Download the AI model (one time, on normal internet)

1. While the phone is on your **normal Wi-Fi** (not the camera's), choose **Qwen3-VL 2B** and tap **Load**. The download is about 1–2 GB and Chrome keeps it afterwards.
2. Wait for "Loaded… Runs fully on this phone now".

7.5 Voice (one time)

1. **Persian voice output:** Settings → General management → Text-to-speech → Preferred engine: **Speech Services by Google** → ⚙ → Install voice data → **Persian**. If Persian isn't offered, install the **sherpa-onnx TTS Engine** app with a Persian Piper voice and make it the preferred engine.
2. **Offline speech input:** Google app → Settings → Voice → Offline speech recognition → download **Persian**.

7.6 Connect to the glasses

1. Plug the glasses cable into the phone (OTG). Wait about 5 s.
2. Wi-Fi → `OpenVision-Cam` → **Keep connection** when Android warns about no internet.
3. In the app: **Settings** → **Camera** → **"ESP32 Wi-Fi camera (glasses)"**, with the address `192.168.4.1`.
4. The pill next to the address should turn green: **"connected · button on"**.

8 · First use and final checks

You do	What should happen
Tap the glasses button	A thumbnail appears, the screen shows "Thinking…", then the phone describes the scene in 1–2 sentences and reads it aloud.
Hold it (≥ 0.5 s)	The picture from the moment you pressed is kept, the microphone opens ("Listening…"), and you ask your question out loud.
Press **again while listening**	The phone stops listening and answers.
Press while it is **talking**	The speech stops and a new picture is taken.

Final checklist

- ☐ The pill shows "connected · button on".
- ☐ The picture is upright, sharp, and aimed where you look (if it's too high or low, change `tilt` and reprint).
- ☐ 10 presses in a row: every tap is described, every hold asks.
- ☐ The arm still folds, and the cable doesn't pull when you turn your head.
- ☐ After 20 minutes the box is warm but comfortable on your temple.
- ☐ **Offline test:** airplane mode on, then Wi-Fi back on to `OpenVision-Cam` only. Reload the page; it should still load and describe. If it doesn't, keep mobile data on (see Troubleshooting).

9 · Updating the firmware over Wi-Fi (pins are cut)

1. Power the glasses camera (phone or any USB charger).
2. Connect the **PC** to the Wi-Fi `OpenVision-Cam`.
3. In Arduino IDE: **Tools** → **Port** → under **Network ports**, choose **openvision-cam at 192.168.4.1**. Keep the same board settings as in 3.3.
4. Click **Upload**. When asked for a password, enter your `AP_PASS`. The board restarts by itself.
5. If the network port doesn't show up: restart Arduino IDE, and allow it through the Windows Firewall when asked.

> If you change `AP_PASS`, the next over-the-air update needs the **old** password, and the phone will need the new one.

10 · Troubleshooting

Problem	Likely cause	Fix
Upload: `Failed to connect`	Board not in download mode	Hold **IO0** on the MB while clicking Upload. Try upload speed 115200, and a different micro-USB cable (it must be a data cable).
No COM port appears	CH340 driver missing	Install CH341SER from wch.cn and replug.
Serial: `JPEG init failed … trying RGB565`	Your camera is a GC2145 / RHYX-M21, not an OV2640	Normal, nothing to do: the firmware switches to software JPEG. Check `http://192.168.4.1/status` shows `"jpeg":"software"`. If the colours look wrong, that camera's colour order is the cause; tell me and I'll add a fix.
Serial: `Camera init failed`	Ribbon loose or crooked	Open the latch, reseat the ribbon straight and fully in, close the latch, press RST.
Serial: `Brownout detector was triggered`, keeps restarting	Not enough power over the thin cable	Check the capacitor is fitted the right way round. Try a shorter or thicker cable. Test with a wall charger; if that works, the cable or OTG adapter is the weak point.
Phone doesn't see `OpenVision-Cam`	No power or not started	Measure 5 V at the 5V/GND pins. Try a different OTG adapter.
`http://192.168.4.1` won't open	Android sends the request over mobile data	Choose "Keep connection" for the camera Wi-Fi. Turn **mobile data off**. Turn off "Switch to mobile data" / "Intelligent Wi-Fi" in Wi-Fi → ⋮ → Advanced / Intelligent Wi-Fi.
Pill: **"blocked – open this page via http://localhost"**	Page opened from an https site	Use Termux and `http://localhost:8080/index.html` (section 7).
Pill: "not connected"	Wrong Wi-Fi or address	Check the phone is on `OpenVision-Cam` and the address is `192.168.4.1`. Reload the page.
Page doesn't load when offline	The AI library comes from the internet and isn't cached	Load the page once on normal internet. If it still fails offline, keep mobile data on and test whether pictures still arrive (it works on many phones). Otherwise switch to hotspot mode (next row).
Hotspot mode (the camera joins the phone)	Keeps the phone's internet working	In the sketch set `USE_AP 0`, `STA_SSID` / `STA_PASS` = your phone's hotspot. Turn the hotspot on. Find the camera's address under Mobile Hotspot → Connected devices (or on Serial), then type it into the app's address field.
Button does nothing	Wiring, or the page isn't connected	The pill must show "connected · button on". Check continuity on the button. Check the wire goes to **IO13**, not IO12.

A tap is treated as a hold	Pressing too long	Raise `LONG_MS` in the sketch (e.g. 600) and update over Wi-Fi.
Picture blurry or milky	Lens film or fingerprint	Remove the protective film and clean the lens. Images are only 384–640 px, so small text needs "High" image detail.
Picture upside down / mirrored	Camera orientation	`FLIP_V` / `MIRROR_H` in the sketch → update over Wi-Fi.
Box too hot	Wi-Fi + camera running all the time	Make sure the vents aren't covered. The sketch already uses 160 MHz and a 10 MHz camera clock. Lower the Wi-Fi power to `WIFI_POWER_8_5dBm`.
Box twists or slides on the arm	Strap too loose, or a smooth metal arm	Tighten the straps. Use rubber (inner tube) as the pad instead of foam. On a very thin metal arm, wrap the arm with 2–3 turns of electrical tape under the pad.
Glasses tip toward the camera side	18 g on one arm	Tape a small weight (a coin or fishing sinkers, about 10 g) to the other arm behind the ear.
Chrome "Aw, Snap!" while thinking	Phone out of memory	Leave auto-recover on. Use image detail "Low" or the Qwen3.5 0.8B model.

11 · Reference card

Connections

From	To
Cable red (+5 V)	capacitor + → **5V** pin
Cable black	capacitor − → **GND** pin
Button leg A	**IO13**
Button leg B (diagonal)	**GND**
Cable green / white	not used (insulate)

Arduino IDE

Board	ESP32 Wrover Module
PSRAM	Enabled
Partition	Minimal SPIFFS (1.9MB APP with OTA)
Upload speed	460800
Serial	115200 baud
OTA port	openvision-cam at 192.168.4.1

Network

Wi-Fi name	`OpenVision-Cam`
Password	your `AP_PASS`
Camera address	`192.168.4.1`
Test page	`http://192.168.4.1/`
Picture	`/capture` · `/held`
Button events	`:81/events`
Status	`/status`

Sketch settings you might change

`AP_PASS`	Wi-Fi + OTA password (8+)
`LONG_MS`	450 ms tap/hold limit
`FLIP_V`, `MIRROR_H`	picture orientation
`JPEG_Q`	12 (10 = sharper, 15 = faster)
`USE_AP`	1 = own Wi-Fi, 0 = join a hotspot

3D model key values

`cam_type`	esp32cam
`tilt`	10° down
`fit_tol`	0.25 (0.35 if tight)
`mount_style`	strap (fits any glasses)
`strap_w`	5.0 mm guide gap
Box outside	≈ 66 × 31 × 13 mm

OpenVision · design files: firmware/openvision_cam/openvision_cam.ino, hardware/glasses_cam_mount.scad, index.html. This build hasn't been tested on real hardware yet. Check each stage (desk test, test-fit, power test) before moving on to the next.